

Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

Ishanne Pophale and Jarek Nowak

Lancaster University

18 August 2026

$\nu_\mu + \bar{\nu}_\mu$ charged-current single charged pion • FHC, RHC, combined

Part I — Inclusive $CC1\mu1\pi^\pm Xp$

1. Motivation & signal
2. Beam, exposure, selection
3. Cut-flow & performance
4. Unfolding & response
5. Differential results + generators
6. Integrated & total cross sections
7. Systematics & sideband validation

Part II — Proton-tagged (W / TKI)

1. Motivation: hadronic mass & TKI
2. Selection & proton-tag performance
3. Observables & closure
4. Generator comparison
5. Background composition

Part III — Multi-pion (2π , 3π)

1. Selection tuning & performance
2. Dedicated pion-ID BDT

Backup

Part I — Inclusive $\text{CC } 1\mu 1\pi^\pm Xp$

Signal definition

$\nu_\mu/\bar{\nu}_\mu$ charged-current, exactly one muon and one charged pion, any number of protons, no π^0 /kaons/heavier mesons; muon $p_\mu > 0.15$, pion $p_\pi > 0.175$ GeV/c, in the fiducial volume.

- Signal defined by the **final state** $\Rightarrow$ every interaction mode contributes, via primary pion production *and* via FSI:

Mode	% of signal	Pion origin
RES ($\Delta(1232)$, higher N^*)	77.5	primary
DIS / SIS	17.5	primary (higher W)
QE	2.6	FSI only (no primary π)
COH	2.1	primary, low $ t $
MEC / 2p2h	0.3	FSI only

- FSI also **migrates** events both ways: absorption/charge exchange turns multi- π into 1π , and 1π into 0π .
- NuMI off-axis: ν_μ -dominant FHC, $\bar{\nu}_\mu$ -enriched RHC. Observables: p_μ , p_π , $\cos \theta_\mu$, $\cos \theta_\pi$, $\theta_{\mu\pi}$.

Exposure (fake data, blind)

FHC (Runs 1,2,4,5)	8.86×10^{20} POT
RHC (Runs 1,2,3,4)	1.11×10^{21} POT
Combined	1.99×10^{21} POT

Event selection: cut sequence

A $CC1\mu1\pi^\pm Xp$ topological selection: one muon, exactly one contained charged pion, any number of protons. Applied in order (cosmic rejection first, then PID, then kinematic cleanup):

1. **NeutrinoSlice** — ≥ 1 track, software trigger.
2. **InFiducialVol** — SCE-corrected ν vertex in FV.
3. **Topological** — Pandora score > 0.67 ; primary cosmic rejection (EXT $\times 33$ down).
4. **MuonCandidate** — ≥ 1 primary track passes μ PID; longest = μ .
5. **ContainedPion** — exactly one other track passes π PID, endpoint contained; highest LLR = π .
- 6–7. **μ/π In3Planes** — hit on each of U, V, Y.
8. **ShowerCut** — veto primary EM showers (π^0 rejection; one soft shower tolerated).
9. **OpeningAngle** — $\theta_{\mu\pi} < 2.6$ rad (149°); rejects large-angle cosmic/dirt.
10. **Nonprotons** (final) — < 4 non-proton-like tracks (LLR > 0.1) and < 5 total primary tracks.

Per-run POT scaling; EXT (beam-off) supplies data-driven cosmics; dirt from the GENIE NuMI overlay. **Final: efficiency** 15.6%, **purity** 55% (measured phase space).

Particle identification (muon & pion candidates)

Track-level PID combines Pandora track score, LLR calorimetric PID, Bragg-peak likelihoods and dedicated MIP/pion BDTs:

Muon candidate (Cut 4)

Variable	Threshold
Track score	> 0.80
Vertex distance	≤ 4.0 cm
Track length	> 10.0 cm
LLR PID	> 0.20
MIP BDT	≥ -0.10

Pion candidate (Cut 5)

Variable	Threshold
LLR PID	> 0.10
Vertex distance	< 4.0 cm
Track length	> 20.0 cm
MIP BDT	> -0.10
Pion BDT (vs. p)	> -0.10
Bragg-pion score	≥ 0.08
Endpoint contained	yes

Momentum reconstruction on the next slide.

Momentum reconstruction: range, MCS and the proton range-KE

All three tracks are **range-based** where possible: a contained track's length maps onto momentum through the particle's dE/dx , which is far more precise than calorimetry at these energies.

Particle	Estimator	PeLEE branch	Note
μ (contained)	range momentum	trk_range_muon_mom_v	37% of selected
μ (exiting)	MCS momentum	trk_mcs_muon_mom_v	63% — range invalid
π	range momentum, μ hypothesis	trk_range_muon_mom_v	m_μ close to m_π
p	range KE , p hypothesis	trk_energy_proton_v	→ momentum, see below

p_μ is therefore a **hybrid** quantity (`..mom_reco` switches on muon containment) and is **MCS-dominated**: NuMI muons are energetic and the TPC is small, so most leave it.

Proton: range KE → momentum

The proton branch gives a *kinetic energy*, so it is converted:

$$|\vec{p}_p| = \sqrt{T_p^2 + 2 T_p m_p}$$

Threshold $|\vec{p}_p| > 0.3 \text{ GeV}/c$ ($T_p \gtrsim 48 \text{ MeV}$) — below this the proton is too short to track. Performance: **−9% bias, 21% resolution**.

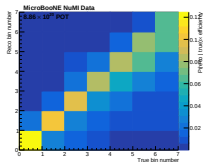
Why the pion uses the μ hypothesis

$m_\mu = 105.7 \text{ MeV}$, $m_\pi = 139.6 \text{ MeV}$: the masses are close, so the range→momentum curve is a good proxy for a stopping pion.

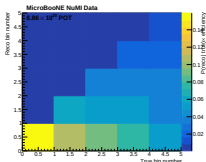
It also returns a **momentum**, directly comparable to the true p_π it is binned against — which keeps the migration matrix near-diagonal.

Valid only for **contained** tracks; the selection requires pion containment.

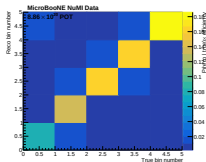
Response (smearceptance) matrices — FHC



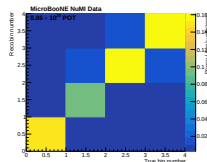
p_μ



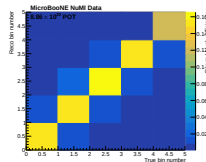
p_π



$\cos \theta_\mu$



$\cos \theta_\pi$



$\theta_{\mu\pi}$

$P(\text{reco} | \text{true}) \times \text{efficiency}$; columns are true bins, rows reco bins. p_μ and the three angular observables are **near-diagonal** — note p_μ is 63% MCS-reconstructed (exiting muons), whose poorer resolution sets its width. p_π is the **outlier** — reco piles into the first bin from every true bin, since pions interact hadronically before stopping so a range-based momentum **under-estimates** p_π . With 50–99% bin migration overall a plain matrix inversion is unusable, hence **Wiener-SVD** regularisation — at the cost of the A_C additional smearing.

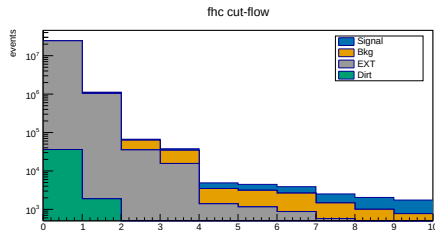
Inclusive cut-flow — FHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	5 580	279 180	24.4M	100.0	0.02
InFiducialVol	4 632	62 146	1.04M	83.0	0.42
Topological	3 501	28 040	35 184	62.7	5.2
MuonCandidate	3 143	18 980	15 516	56.3	8.3
ContainedPion	1 395	2 094	1 376	25.0	28.5
MuonIn3Planes	1 336	1 975	1 145	23.9	29.9
PionIn3Planes	1 239	1 770	872	22.2	31.9
ShowerCut	1 048	898	569	18.8	41.6
OpeningAngle	1 040	813	190	18.6	50.9
Nonprotons	974	643	131	17.5	55.7

MC prediction, per-run POT scaled; data blind. Dirt $\lesssim 1$ event at final cut.

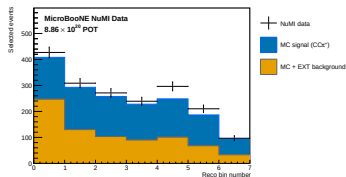
Final cut (Nonprotons), all beams:

Beam	Signal	Bkg	EXT	Eff%	Pur%
FHC	974	643	131	17.5	55.7
RHC	1 081	746	136	17.6	55.0
Comb	2 055	1 389	266	17.5	55.4

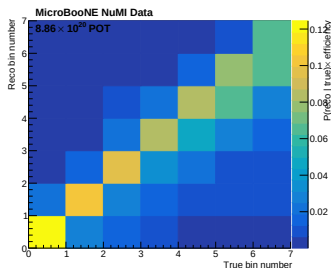


FHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale). Two big drops: Topological (cosmics) and ContainedPion (single- π topology).

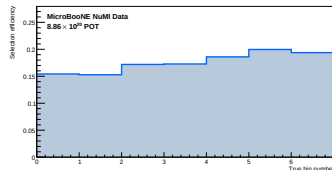
Reconstruction, response & efficiency



Reco p_μ (data/MC stack)



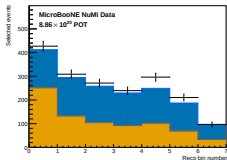
Response matrix



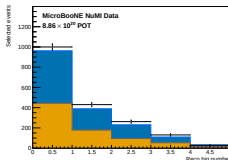
Efficiency vs p_μ

Wiener-SVD unfolding (framework WienerSVDUnfolder, 2nd-derivative regularisation) with the full systematic covariance from the universe throws.

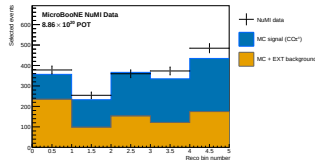
FHC: selected events (POT-scaled), signal vs background



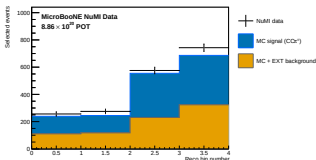
p_μ



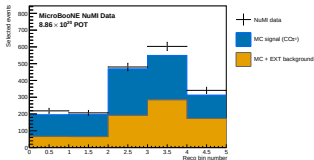
p_π



$\cos \theta_\mu$



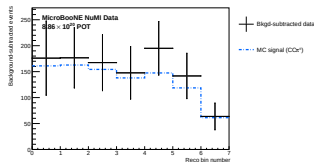
$\cos \theta_\pi$



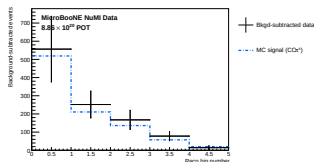
$\theta_{\mu\pi}$

Flat reco-bin space, scaled to full FHC exposure. Points: (blind) fake data; stack: MC **signal** + **background**. Purity $\simeq 55\%$ at the final cut.

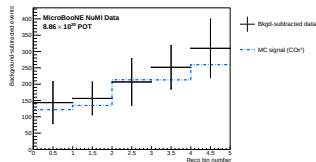
FHC: after background subtraction



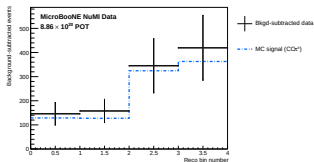
p_μ



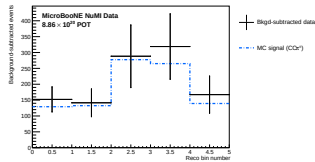
p_π



$\cos \theta_\mu$



$\cos \theta_\pi$

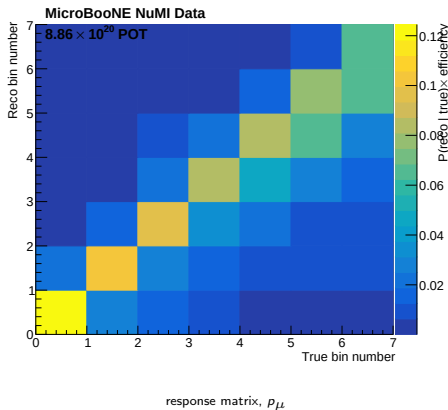


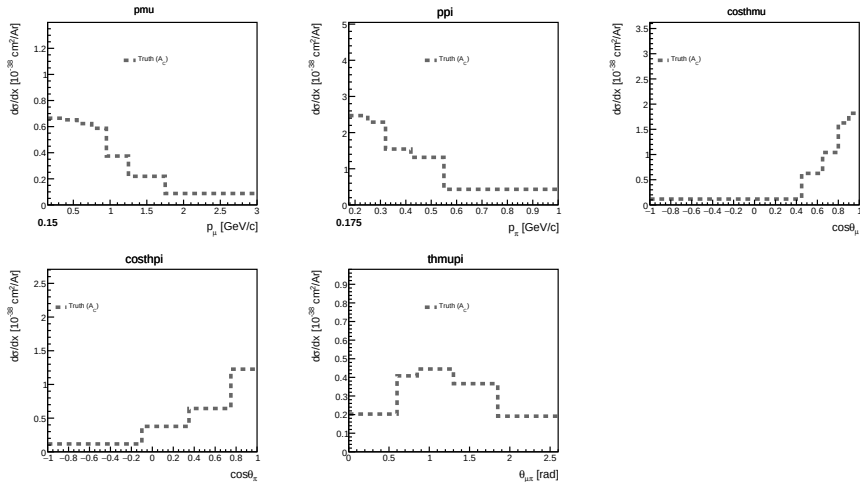
$\theta_{\mu\pi}$

Data — estimated background vs CV MC signal — the input to the unfolding. Background from MC (ν) + beam-off EXT (cosmic) + dirt overlay; validated by the sidebands.

FHC: unfolding

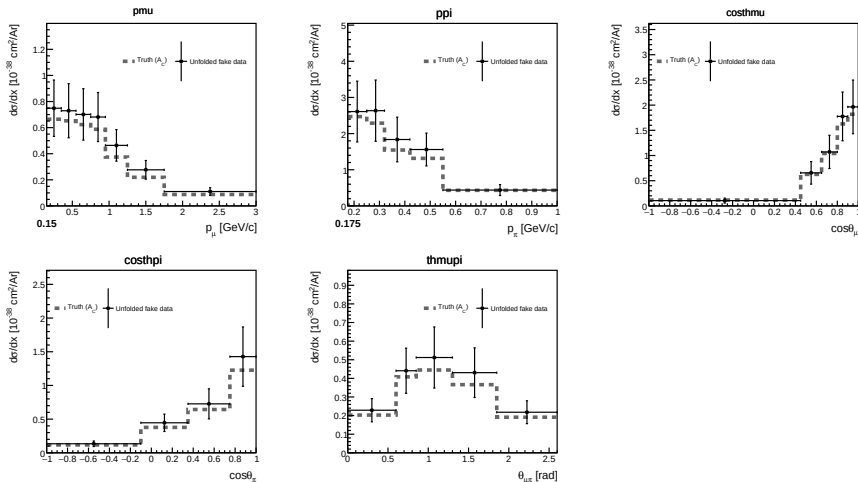
- ▶ The background-subtracted reco spectrum is **not** the cross section: it still carries the detector response and the selection efficiency.
- ▶ Unfolding inverts the response matrix A_{ij} shown earlier. With 50–99% bin migration a plain inversion is unusable, so the analysis uses **Wiener-SVD** regularisation.
- ▶ The price is the **additional-smearing matrix** A_C : the result is $A_C \times (\text{true signal})$, so **every** model must be multiplied by A_C before it is compared to the data.
- ▶ The next slides build up that comparison one curve at a time.





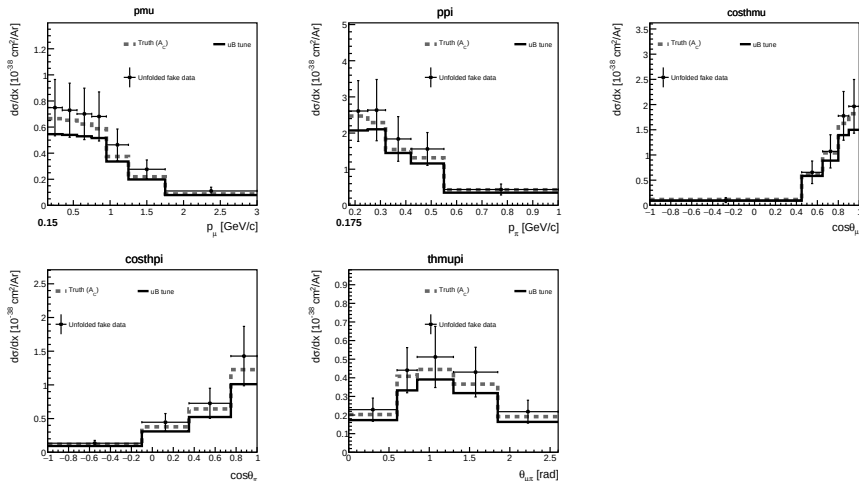
Poisson-thrown fake data are generated from the MicroBooNE tune; after unfolding we should recover this A_C -smeared truth.

Differential cross sections — FHC: + unfolded fake data



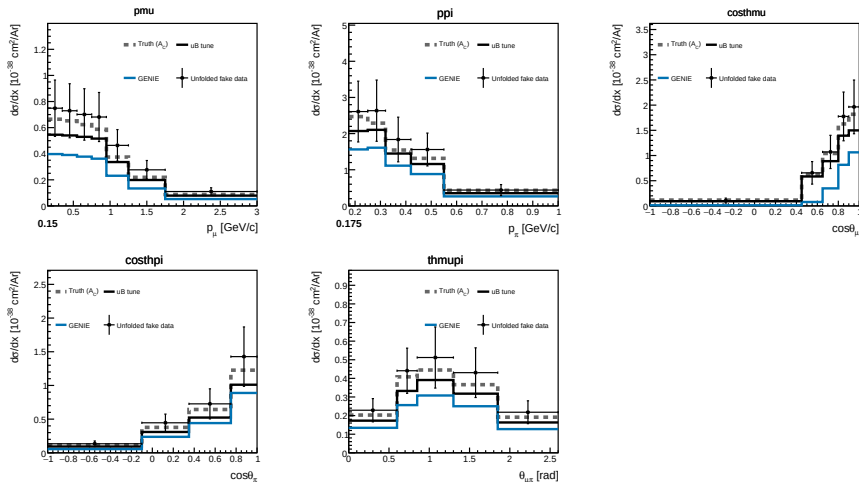
The unfolded points sit on the truth: the response, background subtraction and normalisation close.

Differential cross sections — FHC: + MicroBooNE tune



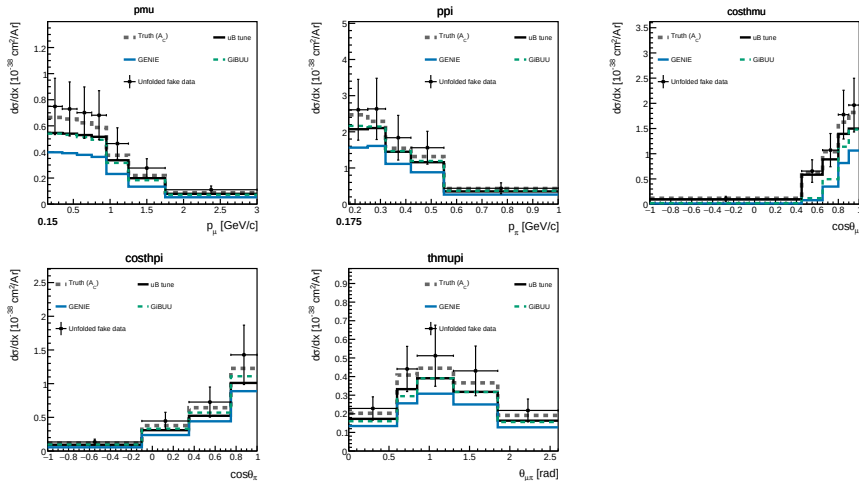
The tune is the central-value model the fake data were thrown from.

Differential cross sections — FHC: + GENIE

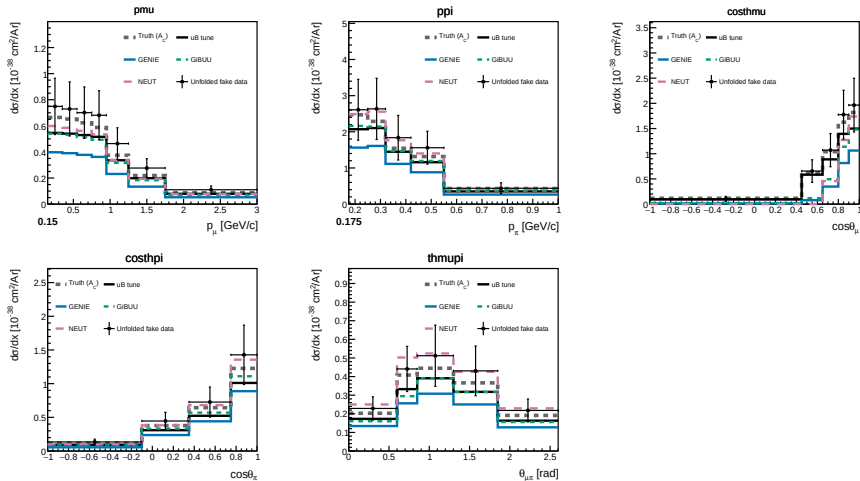


External generators enter, each A_C -smeared so it lives in the same space as the measurement.

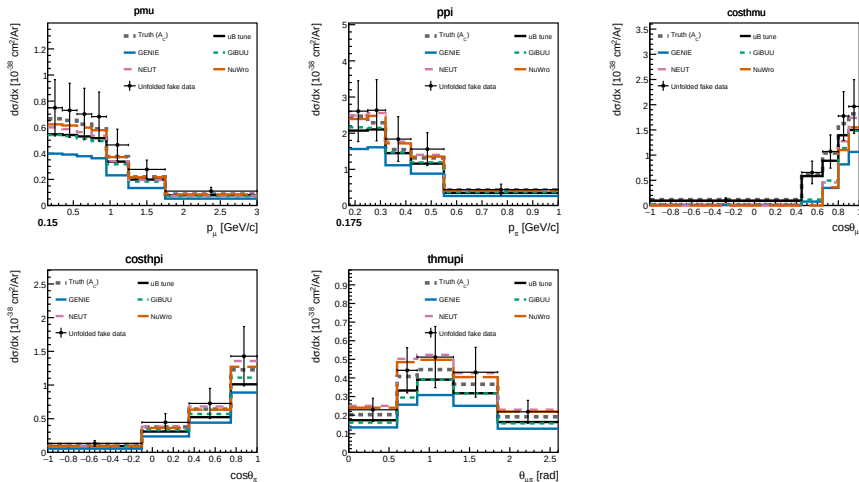
Differential cross sections — FHC: + GiBUU



Differential cross sections — FHC: + NEUT

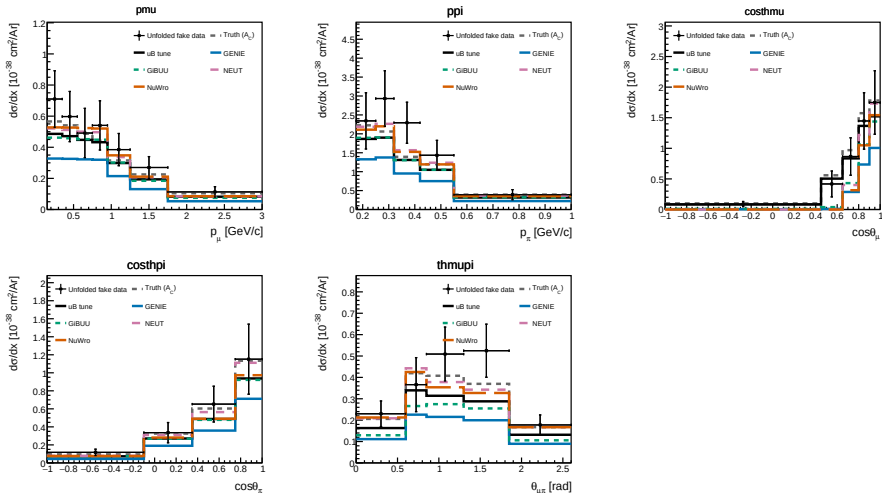


Differential cross sections — FHC: + NuWro



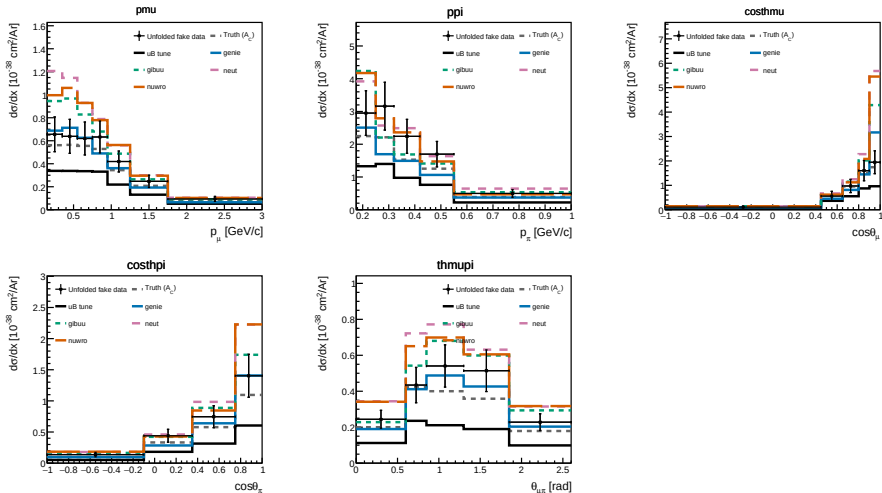
All four generators bracket the data. Error bars are stat $\oplus$ syst: the full covariance (flux, cross-section, detector, re-interaction, POT, targets, data/MC/EXT stats) propagated through the unfolding — $\simeq 27\%$ for p_μ (FHC), flux-dominated (23%).

Differential cross sections — RHC



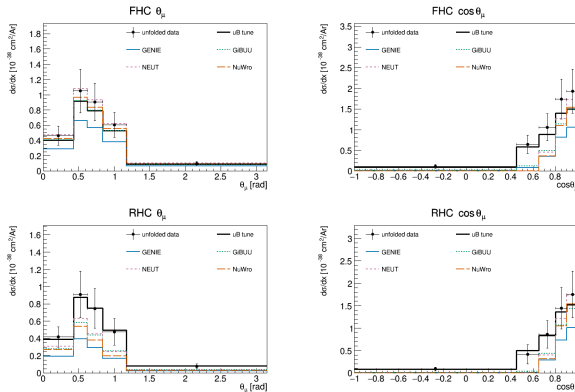
$\bar{\nu}_\mu$ -enriched beam; unfolded fake data vs A_C -smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections — combined



FHC+RHC combined exposure (per-universe correlated flux treatment). Error bars are $\text{stat} \oplus \text{syst}$ (full covariance).

Muon angle: $\cos \theta_\mu$ vs θ_μ



All curves A_C -smeared (same space as the data). Same events, same binning (θ edges = arccos of the $\cos \theta$ edges) — only the variable differs.

Why θ_μ : the A_C smearing penalises the forward peak

Beam	Var	data	tune	GENIE	NuWro
FHC	$\cos \theta_\mu$	0.82	0.67	0.27	0.30
FHC	θ_μ	1.00	0.87	0.63	0.92
RHC	$\cos \theta_\mu$	0.66	0.63	0.23	0.26
RHC	θ_μ	0.83	0.82	0.33	0.42

A_C suppression (smeared/truth):

FHC $\cos \theta_\mu$ 0.33 $\rightarrow$ θ_μ 0.76 (p_μ 0.64)

RHC $\cos \theta_\mu$ 0.34 $\rightarrow$ θ_μ 0.49 (p_μ 0.72)

- ▶ $\cos \theta_\mu$ is sharply forward-peaked; the regularised measurement cannot resolve it, so A_C smears it away — penalising the **peaked generators** far more than the smoother tune.
- ▶ In θ_μ the $\sin \theta$ Jacobian flattens the peak $\Rightarrow$ suppression $0.33 \rightarrow 0.76$, generators land **on** the data.
- ▶ Not a model disagreement: truth-level generator totals agree; it is a **variable-choice artefact**.
- ▶ RHC improves less (poorer angular response, stronger regularisation). $\theta_\pi \approx \cos \theta_\pi$ (pion never sharply peaked).

Integrated & total flux-averaged cross section

Integrated σ_{int} (Wiener-SVD)

	FHC	RHC	Comb
p_μ	0.99	0.86	0.82
p_π	0.96	0.97	0.96
$\cos \theta_\mu$	0.82	0.66	0.70
$\cos \theta_\pi$	0.97	0.81	0.86
$\theta_{\mu\pi}$	0.88	0.88	0.92
θ_μ	1.00	0.83	—
θ_π	0.97	0.80	—

spread = Wiener-SVD A_C smearing ($10^{-38}\text{cm}^2/\text{Ar}$)

Total (cut-and-count, no unfolding)

Config	σ ($10^{-38}\text{cm}^2/\text{Ar}$)
FHC	1.06 ± 0.27
RHC	0.98 ± 0.35
Combined	1.02 ± 0.30

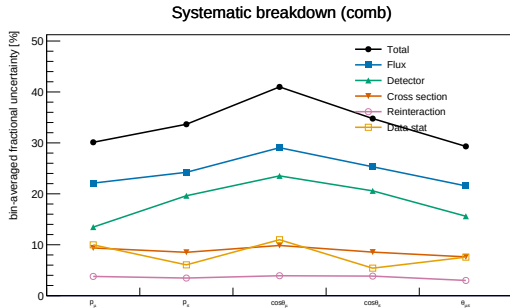
Uncertainties are **syst** $\oplus$ **stat** (e.g. FHC: 25.5% systematic, 4.3% data-statistical). Cut-and-count carries **no** A_C , so it is directly comparable to the truth-level generator column.

Generator totals: the spread is NOT A_C

	truth-level	A_C -smeared	A_C factor
GENIE	0.79	0.51	0.64
GiBUU	1.08	0.69	0.64
NEUT	1.26	0.75	0.59
NuWro	1.20	0.81	0.67

A_C is a **common** $\simeq 0.6$ factor on all four models, so it cannot generate the **$1.6\times$ spread** between them at truth level — that is

Systematics & sideband validation



Flux (PPFX) dominates; detector second; total ~ 30 – 40% .

Background-control sidebands

Four signal-depleted regions ($CC0\pi$, multi- π , π^0 , cosmic). High-statistics data/ ν MC = 1.00 across FHC/RHC/comb $\Rightarrow$ per-run weighting & POT machinery validated; ready to constrain the background at unblinding.

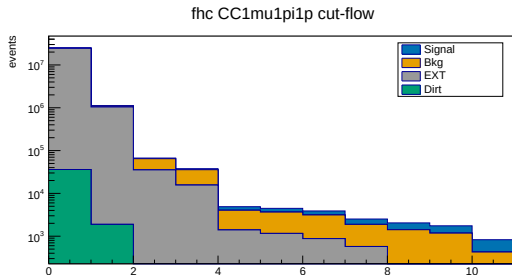
Part II — Proton-tagged subsample: hadronic mass & TKI

Idea

Requiring an identified proton reconstructs the full hadronic system, opening two new observable classes: the **hadronic invariant mass** (probes Δ/N^* directly) and the **transverse kinematic imbalance** (Fermi motion + FSI).

- ▶ CC1 μ 1 π 1 p : inherits the inclusive selection + a leading proton (LLR PID, $p_p \geq 0.3$ GeV/ c , contained). Efficiency 11.5%, purity 48%.
- ▶ Six W/TKI observables: $W_{\pi p}$, W_{had} , δp_T , $\delta \alpha_T$, $\delta \phi_T$, p_n (6 bins each) + five inclusive kinematic observables re-measured on the proton-tagged phase space.

Selection performance & proton-tag quality



Proton tag (truth-matched, FHC)

- ▶ 87% tag purity ($\sim 13\%$ mis-tag)
- ▶ Rejects $\sim 64\%$ background for $\sim 30\%$ signal loss
- ▶ Purity $33\% \rightarrow 48\%$; $S/\sqrt{B} + 17\%$
- ▶ Proton p : -9% bias, 21% resolution

FHC proton-tagged cut-flow; last stage = proton identification

Proton-tagged cut-flow — FHC (full exposure)

Stages None–Nonprotons are the inherited inclusive selection (with the proton-tagged signal definition: inclusive signal + a true proton > 0.3 GeV/c); the final **IdentProton** stage adds the leading-proton identification that defines the subsample.

Cut	Signal	Bkg	EXT	Pred	Eff%	Pur%
None	3 387	281 372	24.4M	24.7M	100.0	0.01
InFiducialVol	2 834	63 944	1.04M	1.11M	83.7	0.26
Topological	2 238	29 303	35 184	66 929	66.1	3.3
MuonCandidate	2 031	20 092	15 516	37 776	60.0	5.4
ContainedPion	852	2 637	1 376	4 890	25.2	17.4
MuonIn3Planes	812	2 499	1 145	4 473	24.0	18.2
PionIn3Planes	754	2 255	872	3 888	22.3	19.4
ShowerCut	632	1 313	569	2 521	18.7	25.1
OpeningAngle	628	1 226	190	2 045	18.5	30.7
Nonprotons	568	1 049	130	1 749	16.8	32.5
IdentProton	401	373	53	828	11.8	48.4

IdentProton rejects $\sim 64\%$ of the ν background while keeping $\sim 71\%$ of signal: purity $32.5\% \rightarrow 48.4\%$. RHC/comb: 46.6%/47.5% purity at $\sim 12\%$ efficiency.

The six proton-tagged observables

Built from the muon, charged-pion and leading-proton four-vectors ($\vec{p}^{\text{had}} = \vec{p}_\pi + \vec{p}_p$, transverse components w.r.t. the beam $\hat{z}$; framework STV tool):

Hadronic mass — probes Δ/N^*

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}$$
$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

$W_{\pi p}$ is **directly measured** — no assumption on the ν direction or energy. W_{had} is calorimetric, $E_{\text{cal}} = E_\mu + E_\pi + T_p$.

Threshold: $W_{\pi p} \geq m_\pi + m_p = 1.08 \text{ GeV}/c^2$.

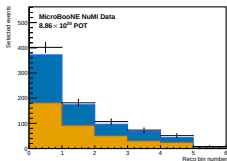
TKI — probes Fermi motion & FSI

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|$$
$$\delta\alpha_T = \angle(-\vec{p}_T^\mu, \vec{p}_T^\mu + \vec{p}_T^{\text{had}})$$
$$\delta\phi_T = \angle(-\vec{p}_T^\mu, \vec{p}_T^{\text{had}})$$
$$p_n = \sqrt{\delta p_T^2 + p_L^2}$$

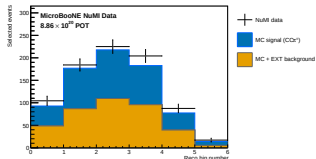
With no nuclear effects $\delta p_T \rightarrow 0$; **Fermi motion broadens** it and **FSI drive $\delta\alpha_T$ toward 180°** . p_n infers the initial-state nucleon momentum ($E_b = 24.78 \text{ MeV}$ argon removal energy enters p_L).

Each is measured in six equal-population bins; the proton is required contained with $|\vec{p}_p| > 0.3 \text{ GeV}/c$.

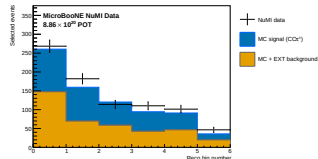
Proton-tagged, FHC: selected events (POT-scaled)



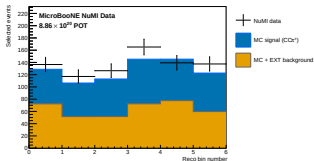
$W_{\pi p}$



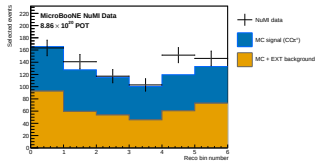
W_{had}



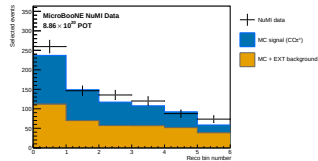
δp_T



$\delta \alpha_T$



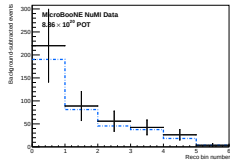
$\delta \phi_T$



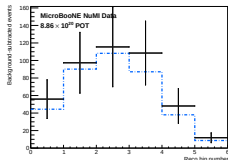
ρ_n

Points: (blind) fake data; stack: MC **signal** + **background**. Purity at the IdentProton stage is 48% — lower than the inclusive selection, so the subtraction is proportionally larger.

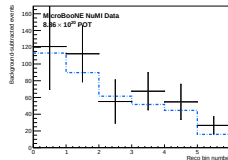
Proton-tagged, FHC: after background subtraction



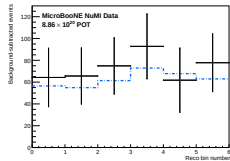
$W_{\pi p}$



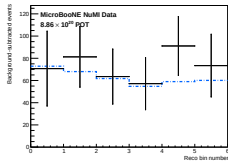
W_{had}



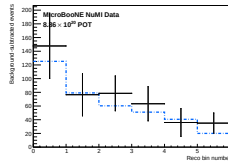
δp_T



$\delta \alpha_T$



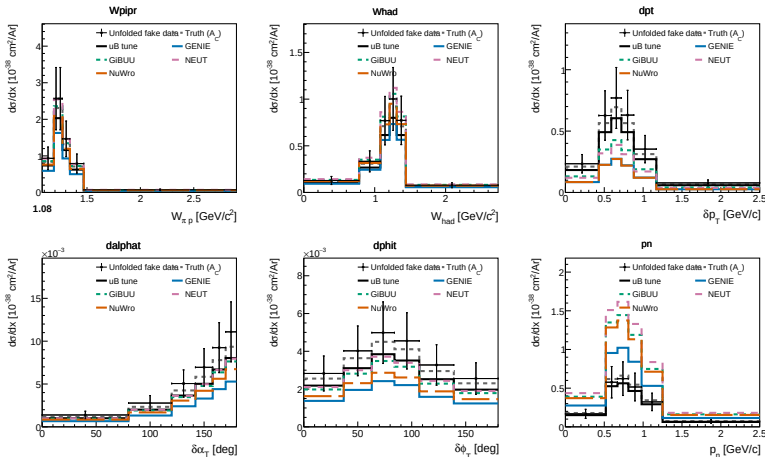
$\delta \phi_T$



ρ_n

Data — estimated background vs CV MC signal — the input to the unfolding for the six W/TKI observables.

Proton-tagged differential cross sections (FHC)



Unfolded fake data vs A_C -smeared truth, the uB tune and all four generators (predictions regenerated at the 6-bin analysis binning). Closure $\chi^2/\text{ndf} \ll 1$ for every observable.

Generator comparison & background composition

Integrated proton-tagged σ ($10^{-38}\text{cm}^2/\text{Ar}$)

Model	σ (truth level)
GENIE	0.50
NuWro	0.65
GiBUU	0.74
NEUT	0.78
Unfolded data ($W_{\pi p}$)	0.62

Spread $\sim 1.6\times$, largest in low- δp_T / high- p_n (FSI-sensitive).
Generator values are truth-level (no A_C); predictions regenerated at the 6-bin analysis binning, NuWro and NEUT from event vectors in the SL7 container.

Remaining background (FHC)

Component	%
ν_μ CC wrong topology	64
True CC1 μ 1 π , no $p>0.3$	18
NC	11
Out-of-FV	5
ν_e CC	1

82% true ν_μ CC — well modelled & constrainable.

Part III — Multi-pion: exclusive $\text{CC } 1_\mu 2\pi$ and $\text{CC } 1_\mu 3\pi$

New exclusive channels, building on the work of **Tusharadri Mahmud** (MSc) and **Linhui Gu**. Thin subselections of the inclusive selection: same muon, but *exactly* 2 or 3 charged pions (any number of protons). They probe the resonant/DIS transition and the intranuclear pion cascade, and are a background to the single-pion signal.

Tuning vs. the 1π selection

Setting	1π	Multi- π
Pion-vertex distance	≤ 4 cm	≤ 9.5 cm (looser)
Uncontained pions	0	up to 2
Opening-angle cut	on	off
Shower veto	on	off
Wire-gap cuts	on	off
Per-pion threshold	0.175	0.10 GeV/c
Pion ID	LLR	dedicated BDT

Performance (Run 1 FHC only)

	2π	3π
Efficiency	17.9%	10.1%
Purity	26.8%	11.3%
Events @ full exposure	~ 670	~ 66
Reco ceiling	66.6%	52.4%

Efficiency/purity measured on the **Run-1 FHC ν_μ overlay alone** (one MC file); event yields are extrapolated to the full FHC exposure. **RHC is not yet processed** for the multi- π channels.

The **reco ceiling** (fraction of true pions reconstructed as tracks at all) shows the loss is **particle-ID**, not reconstruction.

Multi-pion: dedicated pion-ID BDT

- ▶ Inherited loose LLR cut replaced by a **dedicated BDT** (TMVA gradient boost), trained to separate true charged pions from the tracks polluting the multi- π sample. **ROC = 0.81**.
- ▶ Inputs: pion/proton LLR, Bragg p/μ /MIP likelihoods, track score, length, vertex distance — no explicit momentum.
- ▶ 2π uses **two momentum-specialised BDTs** (split at 20 cm track length, tighter soft-pion cut to protect the exact-multiplicity purity); 3π keeps a **single BDT** (its pions are genuinely soft).
- ▶ Trialled on the **inclusive** selection it **hurts** (eff $16.7 \rightarrow 18.9\%$ but purity $60.3 \rightarrow 38.8\%$) $\Rightarrow$ inclusive keeps its LLR cut.

Residual background (truth-matched)

Counted “pion” really is	%
real out-of-topology $\pi^\pm$	44
proton mis-ID	31
muon	12
other	13

Proton contamination is the **irreducible** core — a proton ranging out like a pion is genuinely ambiguous; adding a Bragg veto on top of the BDT did *not* reduce it and only cost efficiency.

Documented in a standalone technote; differential 2π measurement + total σ for 3π are the targets.

Summary

- ▶ **Inclusive:** differential $\text{CC1}\mu 1\pi^\pm$ cross sections for FHC, RHC and combined, per-run POT scaling, Wiener-SVD unfolding; total flux-averaged σ agrees with GENIE/GiBUU/NEUT/NuWro within $\sim 1\sigma$.
- ▶ **Proton-tagged:** new hadronic-mass ($W_{\pi p}$, W_{had}) and TKI (δp_T , $\delta\alpha_T$, $\delta\phi_T$, p_n) observables, plus the five inclusive kinematic observables re-measured. **FHC closure validated** at six-bin granularity; **RHC not yet validated** (unfolds negative — background $>$ data in this low-purity subsample), combined deferred behind it.
- ▶ **Blind analysis:** fake data throughout; sidebands ($\text{CC0}\pi$, multi- π , π^0 , cosmic) validated in all three beams, data/ ν MC $\simeq 1.00$, ready to constrain the background before unblinding.
- ▶ **Multi-pion:** an analysis of the exclusive 2π and 3π channels has started, building on the work of Tusharadri Mahmud (MSc) and Linhui Gu.

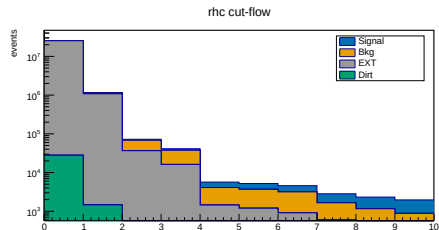
Thank you — questions?

Backup

Backup: inclusive cut-flow — RHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	6 131	304 347	25.3M	100.0	0.02
InFiducialVol	5 124	69 102	1.08M	83.6	0.44
Topological	3 960	31 629	36 532	64.6	5.5
MuonCandidate	3 559	21 443	16 110	58.0	8.6
ContainedPion	1 577	2 649	1 429	25.7	27.8
MuonIn3Planes	1 512	2 500	1 189	24.7	29.0
PionIn3Planes	1 409	2 258	905	23.0	30.8
ShowerCut	1 171	1 061	591	19.1	41.4
OpeningAngle	1 162	965	197	19.0	50.0
Nonprotons	1 081	746	136	17.6	55.0

RHC (Runs 1,2,3,4a-c; 1.108×10^{21} POT). MC prediction, per-run POT scaled; data blind.

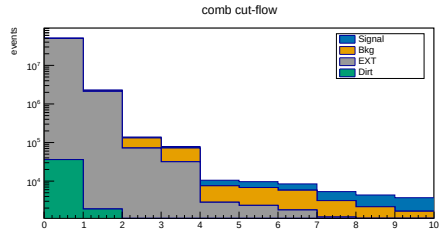


RHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale).

Backup: inclusive cut-flow — combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	11 710	583 527	49.8M	100.0	0.02
InFiducialVol	9 756	131 249	2.12M	83.3	0.43
Topological	7 461	59 669	71 717	63.7	5.4
MuonCandidate	6 702	40 423	31 627	57.2	8.5
ContainedPion	2 972	4 743	2 805	25.4	28.2
MuonIn3Planes	2 847	4 475	2 333	24.3	29.4
PionIn3Planes	2 648	4 028	1 777	22.6	31.3
ShowerCut	2 218	1 959	1 161	18.9	41.5
OpeningAngle	2 202	1 778	387	18.8	50.4
Nonprotons	2 055	1 389	266	17.5	55.4

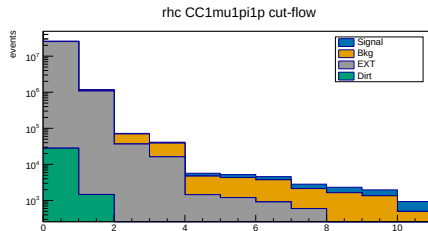
Combined (1.994×10^{21} POT). Efficiency/purity track the two beams closely — the selection is beam-independent.



Combined stack (log scale).

Backup: proton-tagged cut-flow — RHC (full exposure)

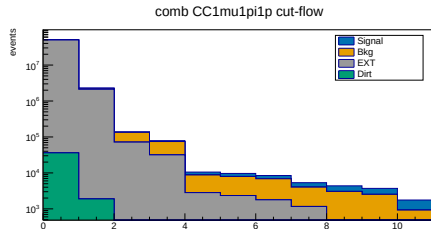
Cut	Signal	Bkg	EXT	Eff%	Pur%
None	3 638	306 840	25.3M	100.0	0.01
InFiducialVol	3 064	71 162	1.08M	84.2	0.26
Topological	2 455	33 134	36 532	67.5	3.4
MuonCandidate	2 234	22 769	16 110	61.4	5.4
ContainedPion	935	3 291	1 429	25.7	16.5
MuonIn3Planes	892	3 119	1 189	24.5	17.1
PionIn3Planes	831	2 835	905	22.9	18.2
ShowerCut	687	1 544	591	18.9	24.3
OpeningAngle	683	1 444	197	18.8	29.4
Nonprotons	611	1 217	136	16.8	31.1
IdentProton	436	443	55	12.0	46.6



RHC proton-tagged stack; last stage IdentProton (leading-proton tag). Purity 31.1% $\rightarrow$ 46.6%.

Backup: proton-tagged cut-flow — combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	7 025	588 212	49.8M	100.0	0.01
InFiducialVol	5 898	135 106	2.12M	84.0	0.26
Topological	4 693	62 436	71 717	66.8	3.4
MuonCandidate	4 265	42 860	31 627	60.7	5.4
ContainedPion	1 787	5 928	2 805	25.4	16.9
MuonIn3Planes	1 704	5 618	2 333	24.3	17.6
PionIn3Planes	1 585	5 091	1 777	22.6	18.7
ShowerCut	1 320	2 857	1 161	18.8	24.7
OpeningAngle	1 310	2 670	387	18.7	30.0
Nonprotons	1 179	2 266	266	16.8	31.8
IdentProton	837	816	109	11.9	47.5



Combined proton-tagged stack; IdentProton purity

31.8% $\rightarrow$ 47.5%.

Backup: per-run exposure

Mode	Run	Data POT	MC/data scale
FHC	1	3.28×10^{20}	0.141
	2	1.27×10^{20}	0.051
	4	2.08×10^{20}	0.073
	5	2.23×10^{20}	0.116
RHC	1–4	1.11×10^{21}	0.045–0.091

Each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$ (real-data-ready).

Backup: systematic breakdown (combined)

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Total	30.1	33.7	41.0	34.8	29.3
Flux (PPFX)	22.1	24.2	29.1	25.3	21.6
Detector	13.5	19.6	23.5	20.6	15.6
Cross section	9.4	8.5	9.9	8.6	7.6
Data stat	10.0	6.1	11.0	5.4	7.6

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

Backup: history — the superseded pion-momentum estimator

An earlier version of this analysis reconstructed the **pion** momentum from the **proton-hypothesis range kinetic energy** (`trk_energy_proton_v`) rather than the muon-hypothesis range momentum used now.

- ▶ That is a **kinetic energy**, while the true axis it was binned against (with identical bin edges) is a **momentum** — two different physical quantities.
- ▶ Consequence: the migration matrix was **systematically shifted** rather than near-diagonal. A true 0.2 GeV/c pion gives $KE \approx 0.104$ GeV, **two bins low**; a true 0.175 GeV/c pion falls **below the first reco bin entirely**.
- ▶ Fixed by switching to `trk_range_muon_mom_v` (`CC1mu1piXp.cxx`), which returns a momentum and exploits $m_\mu \approx m_\pi$.

Note: the proton-hypothesis range KE remains the *correct* estimator for the **proton** in the W/TKI subsample — there the quantity wanted really is a kinetic energy, converted to momentum via $|\vec{p}_p| = \sqrt{T_p^2 + 2T_p m_p}$.

Resolution panels produced before the switch quote a $\sim 21\%$ underestimate of p_π ; they are to be regenerated with the current estimator.

Backup: from PeLEE branches to the kinematic inputs

Every observable is built from three reconstructed tracks (muon, charged pion, leading proton), each taken from the PeLEE NeutrinoSelectionFilter ntuple at the candidate's PFP index:

Quantity	PeLEE branch	Note
$ \vec{p}_\mu $	trk_range_muon_mom_v	range-based (primary)
$ \vec{p}_\mu $ (x-check)	trk_mcs_muon_mom_v	multiple-scattering
$ \vec{p}_\pi $	trk_range_muon_mom_v	μ -hypothesis range, at the π index
$T_p \rightarrow \vec{p}_p $	trk_energy_proton_v	p -hypothesis range KE; $ \vec{p}_p = \sqrt{T_p^2 + 2T_p m_p}$
$\hat{u}_\mu, \hat{u}_\pi, \hat{u}_p$	trk_dir_x/y/z_v	unit track direction
$\cos \theta$	from $\hat{u}_z$	w.r.t. detector z (beam-ish) axis
$\theta_{\mu\pi}$	$\arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$	opening angle
PID / candidacy	trk_llr_pid_score_v, trk_bragg*_v, trk_score_v, trk_distance_v, trk_len_v	see PID slide

Energies follow as $E_i = \sqrt{|\vec{p}_i|^2 + m_i^2}$ with the PDG mass of the assigned species. Truth counterparts use mc_nu_daughter_p[xyz] (back-matched via backtracked_pdg).

Backup: W/TKI observable definitions

From the muon, charged-pion and leading-proton four-vectors above (framework STV tool):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad (\text{direct, no } \nu \text{ direction})$$

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2} \quad (\text{calorimetric})$$

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T, \delta\phi_T \text{ (TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

with $\vec{p}^{\text{had}} = \vec{p}_\pi + \vec{p}_p$, $E_{\text{cal}} = E_\mu + E_\pi + T_p$, and the transverse components taken w.r.t. the beam ($\hat{z}$) axis.
 $E_b = 24.78$ MeV argon removal energy enters p_L (hence p_n).

Backup: proton-tagged binning & closure

Observable	Bin edges (6 equal-population bins)
$W_{\pi p}$	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.00 GeV/ c^2
W_{had}	0, 0.78, 1.08, 1.21, 1.31, 1.44, 1.90 GeV/ c^2
δp_T	0, 0.43, 0.59, 0.72, 0.88, 1.17, 1.60 GeV/ c
$\delta\alpha_T$	0, 80, 120, 142, 158, 170, 180 deg
$\delta\phi_T$	0, 37, 63, 84, 107, 139, 180 deg
p_n	0, 0.52, 0.67, 0.81, 0.98, 1.25, 1.70 GeV/ c

+ five kinematic observables (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) on inclusive binning.

FHC W/TKI closure σ_{int} : 0.62, 0.60, 0.61, 0.67, 0.63, 0.52; $\chi^2/\text{ndf} \ll 1$, all positive.

RHC not yet validated: all 11 observables unfold negative (-0.48 to -0.04) — background $>$ data in this $\sim 47\%$ -purity subsample. Earlier positive RHC values came from an incorrect MC POT normalisation and are retracted. Comb deferred behind RHC.

A_C **smearing factor** (smeared/truth): W_{had} 0.88, $W_{\pi p}$ 0.78 (sharpest distribution, yet barely affected — **no W reparameterisation needed**), $\delta\alpha_T$ 0.63, $\delta\phi_T$ 0.61, **δp_T 0.43**, **p_n 1.71 (enhanced)**. The last two distort model comparison in those panels.

Backup: χ^2/ndf — inclusive, all observables and beams

Unfolded fake data compared with each A_C -smeared prediction.

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	1.66/7	3.09/7	5.99/7	3.87/7	3.77/7	2.90/7
FHC	p_π	1.96/5	3.77/5	5.05/5	3.78/5	3.33/5	3.46/5
FHC	$\cos \theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	$\cos \theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	$\theta_{\mu\pi}$	0.59/5	2.59/5	4.12/5	3.24/5	2.47/5	2.29/5
RHC	p_μ	5.27/7	8.57/7	10.88/7	8.81/7	8.15/7	8.48/7
RHC	p_π	6.09/5	7.04/5	9.36/5	7.28/5	5.96/5	6.13/5
RHC	$\cos \theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	$\cos \theta_\pi$	1.47/4	4.00/4	6.64/4	3.92/4	4.21/4	4.30/4
RHC	$\theta_{\mu\pi}$	4.32/5	5.57/5	7.67/5	5.68/5	6.56/5	7.59/5
Comb	p_μ	3.41/7	3.70/7	5.82/7	3.99/7	3.83/7	3.30/7
Comb	p_π	1.32/5	2.32/5	3.85/5	2.65/5	1.82/5	2.05/5
Comb	$\cos \theta_\mu$	2.49/5	4.61/5	9.07/5	12.02/5	26.72/5	27.20/5
Comb	$\cos \theta_\pi$	0.42/4	1.83/4	4.01/4	1.69/4	1.89/4	1.95/4
Comb	$\theta_{\mu\pi}$	2.10/5	2.42/5	3.91/5	3.05/5	3.49/5	3.45/5

The small *truth* column is the closure. RHC closure is visibly worse than FHC. The $\cos \theta_\mu$ row is an outlier for NEUT/NuWro — see next slide.

Backup: the $\cos \theta_\mu \chi^2$ is largely an A_C artefact

Same events, same data, only the angular variable changes:

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos \theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	θ_μ	0.05/5	3.07/5	4.73/5	3.03/5	2.82/5	2.90/5
RHC	$\cos \theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	θ_μ	0.48/5	4.41/5	6.97/5	5.54/5	8.18/5	8.46/5
FHC	$\cos \theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	θ_π	0.44/4	1.15/4	3.77/4	0.93/4	1.30/4	1.18/4

In $\cos \theta_\mu$ the A_C smearing suppresses the peaked generators by $\times 0.33$ but the smoother tune by much less, so NEUT and NuWro are pushed far from the data and χ^2 explodes. In θ_μ , where the suppression is $\times 0.76$, the **same models agree well** (χ^2 falls by $\sim 9\times$). The π angle, never sharply peaked, is unchanged. **The large $\cos \theta_\mu \chi^2$ is therefore mostly a variable-choice artefact, not model disagreement** — a further argument for quoting θ rather than $\cos \theta$.

Backup: χ^2/ndf — proton-tagged (FHC)

Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro	A_C
W_{had}	0.09/6	1.39/6	1.65/6	1.06/6	1.17/6	1.05/6	0.88
$W_{\pi p}$	0.09/6	0.95/6	1.78/6	0.62/6	0.56/6	0.87/6	0.78
$\delta\alpha_T$	0.26/6	2.74/6	4.73/6	2.95/6	2.78/6	3.53/6	0.63
$\delta\phi_T$	0.09/6	5.14/6	7.21/6	5.61/6	5.39/6	6.47/6	0.61
δp_T	0.46/6	5.34/6	9.89/6	8.27/6	9.02/6	10.38/6	0.43
p_n	0.09/6	4.24/6	7.72/6	18.74/6	25.40/6	16.45/6	1.71

Rows are ordered by how close the additional-smearing factor A_C is to unity, and the χ^2 follows it **monotonically**: $|A_C - 1| = 0.12, 0.22, 0.37, 0.39, 0.57, 0.71$ against generator $\chi^2 \simeq 1.2, 1.0, 3.3, 6.2, 9.4, 17$. The two observables A_C barely touches ($W_{\text{had}}, W_{\pi p}$) agree with **all four** generators; the two it distorts most (δp_T suppressed to 0.43, p_n *enhanced* to 1.71) show the largest χ^2 — in the same direction for every model. **This is the signature of a smearing artefact, not model discrimination**, the same effect seen in $\cos\theta_\mu$. The closure (truth column) is excellent throughout.

Backup: generator predictions

- ▶ Inclusive: per-observable $d\sigma/dx$ from GENIE (gst), GiBUU (FinalEvents), NEUT (neutvect), NuWro (event output), flux-combined to per-argon 10^{-38} cm^2 .
- ▶ Proton-tagged W/TKI: same event samples reprocessed with the proton tag + TKI construction; NuWro and NEUT run inside the SL7 aptainer container.
- ▶ FTE rebinned to the coarsened analysis binning so every generator overlays on every observable.